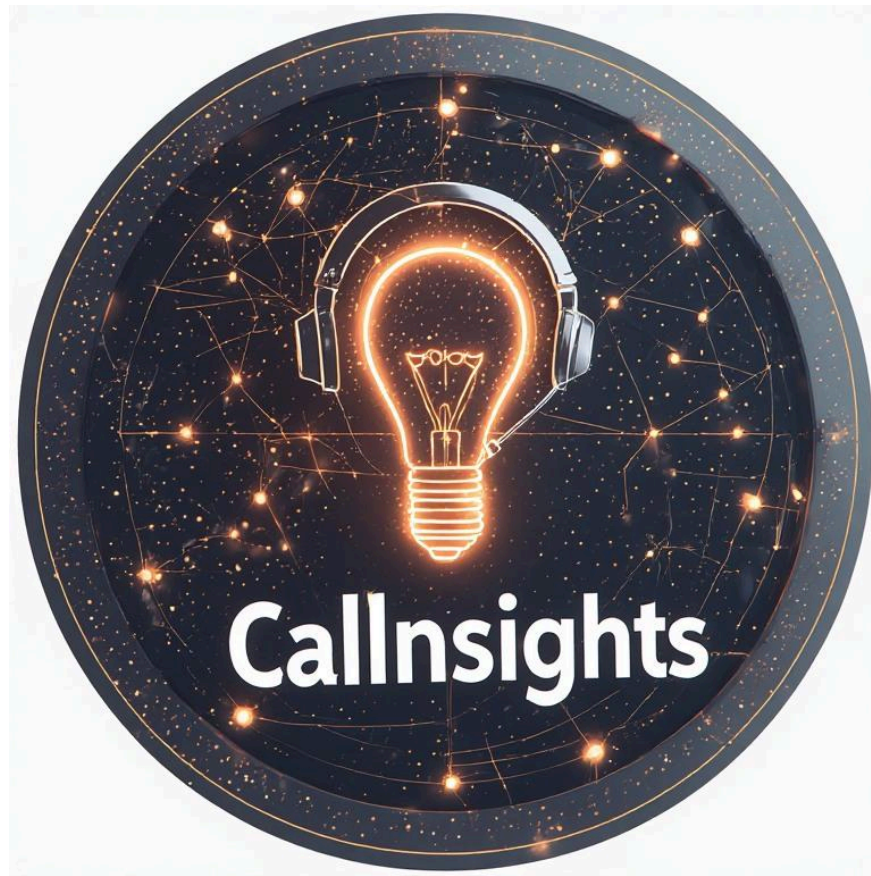




Senior Capstone: Design Document

Call Insights (CI): A Call Transcript Analysis Framework

Team Members: Abdulrahman Al Hattali, Alex Held, Anshul Kasera, and Prachi Aswani

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1. Executive Summary

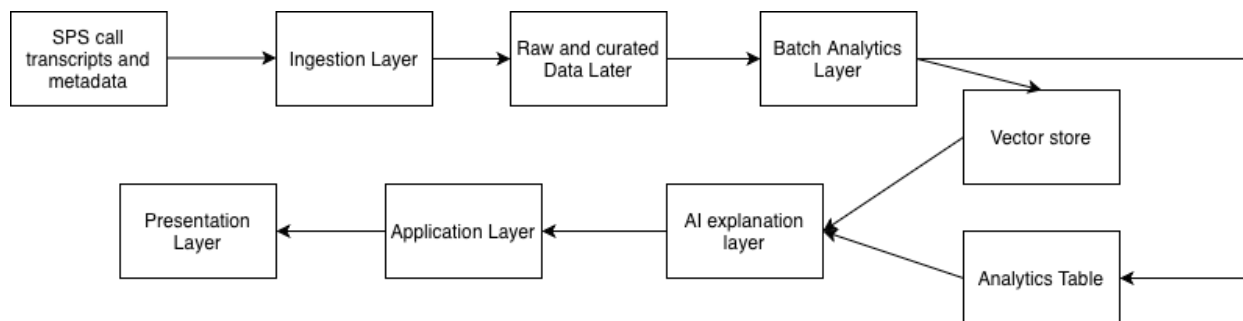
Customer service call centers generate vast amounts of conversational data, yet most organizations are only able to analyze a small fraction of it due to time, staffing, and tooling constraints. As a result, important insights about customer needs, agent effectiveness, and operational inefficiencies often go unnoticed. This project proposes Call Insights (CI), a modular analytics framework that transforms raw customer service call transcripts into actionable, evidence-based insights which will be clearly relayed over several dashboards for operational decision-makers.

CI is designed as a batch-driven call analysis system that processes historical call transcripts to identify recurring customer issues, agent behaviors, and call outcomes. Unlike traditional quality assurance approaches that rely on manual review or surface-level metrics, CI analyzes calls at scale and recognizes that a single call may involve multiple issues, multiple agent strategies, and mixed outcomes. By segmenting calls into issue-specific components, CI enables more precise analysis of what happens during customer interactions.

The system is primarily intended for call center managers, operations analysts, and strategy teams who need reliable insight into which agent behaviors lead to positive outcomes and which customer issues remain unresolved or are resolved inconsistently. CI correlates customer intent, agent actions, and outcomes across thousands of calls to surface statistically supported patterns. These patterns are presented alongside transparent traceable evidence as to how they were reached, allowing stakeholders to understand not only *what* insights are produced, but *why* they are trustworthy.

A central goal of CI is to support closed-loop learning. Insights generated by the system can inform operational strategies, such as changes to agent training or call handling guidelines, which are then tracked over time to evaluate their effectiveness. This feedback loop allows organizations to move beyond anecdotal decision-making toward continuous, data-driven improvement.

CI is being developed as a secure, self-contained framework suitable for enterprise environments, with no reliance on external APIs or live internet access. This allows the system to work with private customer data, ensuring customer data won't be leaked. By prioritizing modularity, interpretability, and evidence-first analysis, CI aims to provide a practical and extensible foundation for large-scale call intelligence rather than a single-purpose analytics tool.



2. Background

Overview of the Idea Space

Customer service organizations rely heavily on phone calls to resolve customer issues, yet these interactions are difficult to analyze at scale. Manual call reviews are expensive and inconsistent, while traditional metrics such as call duration or resolution codes often fail to capture nuance in what actually happened during the conversation. As a result, organizations struggle to answer key questions such as why customers call, which agent behaviors are effective, and which issues lead to repeat calls.

The CI framework is motivated by the need to extract structured, actionable information from unstructured call transcripts. Rather than treating calls as single, monolithic events, CI models calls as sequences of issue-specific interactions, each with its own intent, behavior, and outcome. This approach enables more accurate analysis of complex calls and supports a deeper understanding of customer experience and agent performance.

The primary users of CI include agent managers, call center leadership, quality assurance teams, and operational analysts within enterprise service organizations such as SPS. These users require insights that are scalable, explainable, and directly tied to measurable outcomes. CI addresses this need by combining large-scale transcript processing with transparent analytical methods and clearly documented evidence.

Similar Ideas

Several commercial platforms offer call analytics and conversation intelligence capabilities. These systems typically provide sentiment analysis, keyword detection, or agent scoring dashboards. While effective for high-level monitoring, many existing solutions function as black-box systems, offering limited visibility into how insights are generated or how they relate to long-term outcomes.

In contrast, CI emphasizes:

- Explicit modeling of customer issues, agent behaviors, and outcomes
- Evidence-backed insights with traceable call examples
- Support for longitudinal analysis and strategy evaluation
- Deployment within data privacy focused, enterprise-controlled environments

Rather than competing as a full-scale commercial product, CI is designed to function as a framework that demonstrates how call intelligence can be implemented in a principled, extensible, and auditable way.

Feature	Traditional Call Analytics	Call Insights (CI) Framework
Analysis Model	Monolithic: Treats a call as a single event with one sentiment or outcome score.	Multi-Pattern: Segments calls into specific issues, each with its own intent, behavior, and outcome.
Transparency	Black-Box: Provides scores (e.g., "Agent Score: 8/10") without clear derivation.	Evidence-First: Every insight is backed by an "Evidence Pack" with traceable transcript examples and IDs.
Logic Engine	Surface-Level: Primarily relies on keyword detection or basic sentiment.	Triple Engine: Correlates specific Customer Problems + Agent Behaviors + Outcomes at scale.

Security	Cloud-Dependent: Often requires external APIs or internet-facing platforms.	SPS-Contained: Fully hosted on-premise; no external APIs or internet access to ensure data privacy.
Operational Impact	Passive Monitoring: Provides retrospective dashboards for high-level monitoring.	Closed-Loop Learning: Directly tracks the adoption and effectiveness of new operational strategies over time.
Architecture	Proprietary/Rigid: Difficult to customize or add new detection logic.	Modular/Pluggable: Allows SPS to add or modify analysis components (NLP/ML) without rebuilding the system.

Required Technology

To achieve its goals, CI relies on a combination of established software technologies:

- **Natural Language Processing (NLP):**
Used for transcript segmentation, topic identification, sentiment analysis, and embedding generation.
- **Machine Learning & Statistical Analysis:**
Clustering, classification, and correlation analysis are used to identify recurring patterns and measure relationships between behaviors and outcomes.
- **Data Processing Pipelines:**
Batch processing pipelines handle large volumes of transcripts and intermediate analytical artifacts.
- **Backend Services & Storage:**
Databases and file storage systems maintain transcripts, derived features, and evidence metadata.
- **Visualization & Reporting Tools:**
Dashboards and reports present insights, trends, and supporting evidence to end users.

The project primarily leverages technologies and skills already familiar to the team, minimizing the need for extensive new learning during the development phase.

Assets and Engines

CI will be built primarily from scratch as a custom analytics framework, while leveraging open-source libraries for NLP, machine learning, and data processing. Pre-existing call transcripts and metadata provided by the partner organization serve as the primary dataset for development and evaluation.

Key reusable assets include:

- Open-source NLP and ML libraries
- Existing transcript storage formats
- Internal call metadata for grounding and validation

No proprietary third-party analytics engines are required, ensuring flexibility and transparency in system design.

Software and Hardware Requirements

Developer / Software Requirements

- A modern development environment with support for Python and data science libraries
- Closed ecosystem, no internet dependent APIs / libraries to ensure data privacy
- Access to anonymized call transcripts for development and testing
- Using SPS's existing stack / workflow where applicable for easier integration

User Requirements

- Access to CI dashboards and reports through a standard web browser
- No specialized hardware beyond typical enterprise workstations for users

Hardware Requirements

- Access to compute resources for running batch processing (provided by SPS)

3. USE CASES

Team Name: CallInsights

Team Members: Abdulrahman Al Hattali, Alex Held, Anshul Kasera, and Prachi Aswani

Use Case 1

Number: UC-1

Title: View Executive Overview Dashboard

Preparer: Prachi Aswani

Actor/User: CTO

User Story

As a CTO, I want to quickly understand the overall health of call center performance so that I can identify risks, trends, and areas needing strategic attention.

Course of Events

1. User logs into CI.
2. System displays Executive Overview Dashboard.
3. User sees:
 - Total call volume (selected time window)
 - % Resolved vs Unresolved
 - Escalation rates
 - Call issue distributions
 - Recurring issues
 - Any other immediate major problems/issues
4. User selects a time filter (e.g., Last 30 Days).
5. Dashboard updates dynamically.
6. User clicks on a top issue to drill deeper (redirects to UC-4).

Exceptions / Alternatives

- If no data exists for selected time window, show "Insufficient Data" state.
- If user lacks permissions, limited metrics view.

Related UI

See UI-1: Executive Overview Dashboard

Use Case 2

Number: UC-2

Title: Analyze Daily Insights Pulse

Preparer: Prachi Aswani

Actor/User: Operations Analyst

User Story

As an operations analyst, I want to see newly emerging issues or spikes in call patterns so that I can respond quickly to operational disruptions.

Course of Events

1. User navigates to "Daily Insights Pulse".
2. System displays:
 - Emerging issue clusters
 - Volume spikes
 - Sentiment shifts
 - Escalation anomalies
3. User clicks an emerging issue.
4. System shows:
 - Trend graph
 - Sample call IDs
 - Sentiment trajectory distribution

5. User exports insight summary.

Exceptions

- No statistically significant changes, show “No significant anomalies detected.”

Related UI

See UI-2: Daily Insights Pulse View

Use Case 3

Number: UC-3

Title: Explore Issue Patterns (Triple Engine Analysis)

Preparer: Alex Held

Actor/User: Strategy Team Member / General Agent Manager

User Story

As a strategy team member, I want to analyze which agent behaviors correlate with positive or negative outcomes for a specific issue so that I can refine agent training strategies.

Course of Events

1. User selects an issue (e.g., “Payment Confusion”).
2. System runs Triple Engine view:
 - Problem
 - Agent Behavior
 - Outcome
3. System displays:
 - Correlation strength
 - Lift
 - Statistical significance

4. User filters by:
 - Agent group
 - Time window
 - Escalation status
5. System updates performance comparison.
6. User clicks “Generate Evidence Pack.”

Exceptions

- Low sample size, system displays statistical warning.

Related UI

See UI-3: Issue Explorer

See UI-5: Strategy Effectiveness Dashboard

Use Case 4

Number: UC-4

Title: Drill Down into Specific Call

Preparer: Alex Held

Actor/User: QA Manager

User Story

As a QA manager, I want to inspect a specific call transcript and its extracted metadata so that I can verify the validity of insights.

Course of Events

1. User clicks on call ID from dashboard.

2. System opens Call Drilldown view.
3. User sees:
 - Full transcript (masked sensitive info)
 - Segmented chunks
 - Issue tags
 - Sentiment over time graph
 - Agent behavior markers
 - Resolution classification
4. User downloads transcript summary.

Exceptions

- If transcript missing, show “Transcript unavailable.”

Related UI

See UI-4: Call Drilldown View

Use Case 5

Number: UC-5

Title: Evaluate Strategy Effectiveness Over Time

Preparer: Abdulrahman Al Hattali

Actor/User: Director of Training

User Story

As a training director, I want to measure whether an accepted strategy improved outcomes over time so that I can justify continued adoption, revision, or retirement.

Course of Events

1. User navigates to Strategy Management.
2. User selects an accepted strategy (for example, “Explain Timeline Early”).
3. User clicks View Effectiveness.

4. System opens the Strategy Effectiveness Dashboard.
5. System displays:
 - Before/After comparison
 - Resolution rate change
 - Average Handle Time (AHT) change
 - First Call Resolution (FCR) change
 - Escalation rate change
 - Sentiment improvement
 - Trend over time / drift indicator
6. User reviews supporting insights and representative calls.
7. User optionally exports a summary or opens the linked Evidence Pack.

Exceptions

- Insufficient “before” data, show limited comparison.

Related UI

See UI-6: Strategy Management View

See UI-5: Strategy Effectiveness Dashboard

Use Case 6

Number: UC-6

Title: Export Evidence Pack Report

Preparer: Abdulrahman Al Hattali

Actor/User: Operations Executive (or QA/Operations Lead)

User Story

As an executive, I want to export an evidence-backed report so that I can share leadership-ready findings and justify recommendations with traceable call evidence.

Course of Events

1. User navigates to Reports & Export (UI-7).
2. User selects a time window for the report.
3. User applies filters (e.g., issue type, team/queue).

4. User selects report type Evidence Pack and clicks Generate Evidence Pack.
5. System compiles an **Evidence Pack artifact** that includes:
 - Time window definition
 - **Filters applied** (queue/issue/team)
 - Counts & trends with key operational metrics (AHT, FCR, escalation rate, sentiment)
 - Sample size
 - Representative call IDs (traceable examples supporting the claims)
 - Derived signals where relevant (e.g., sentiment shift, recurrence)
6. User chooses an export format (PDF or CSV) and downloads the report.

Exceptions/Alternates:

- If there is insufficient data for the selected filters/time window, the system warns the user and suggests expanding the time window or broadening filters.

Related UI

See UI-7: Reports & Export View

Use Case 7

Number: UC-7

Title: Monitor Compliance in Customer Calls

Preparer: Anshul Kasera

Actor/User: Compliance Officer

User Story

As a compliance officer, I want to automatically detect whether agents follow required regulatory scripts so that the organization can ensure compliance with legal and company policies.

Course of Events

1. User navigates to Compliance Monitoring Dashboard.
2. System analyzes call transcripts for required compliance phrases and disclosures.
3. System displays:
 - Percentage of calls meeting compliance requirements

- Calls missing mandatory disclosures
 - Compliance trend over time
- 4. The user selects a non-compliant call category.
- 5. The system displays example call transcripts with highlighted missing phrases.
- 6. User exports a compliance audit report.

Related UI

See UI-8: Compliance Dashboard

Use Case 8

Number: UC-8

Title: Reconstruct Customer Issue Journey

Preparer: Anshul Kasera

Actor/User: Customer Experience Manager

User Story

As a customer experience manager, I want to trace multiple calls from the same customer related to the same issue so that I can understand the full customer journey and identify service breakdowns.

Course of Events

1. The user navigates to Customer Journey Explorer.
2. User searches by customer ID or issue tag.
3. System groups related calls into a timeline view.
4. System displays:
 - Sequence of calls
 - Issue classification changes
 - Sentiment progression
 - Escalation events
5. The user identifies where the issue was first reported and when it was resolved.
6. The user exports the journey summary for customer experience analysis.

Related UI

See UI-9: Customer Journey Explorer

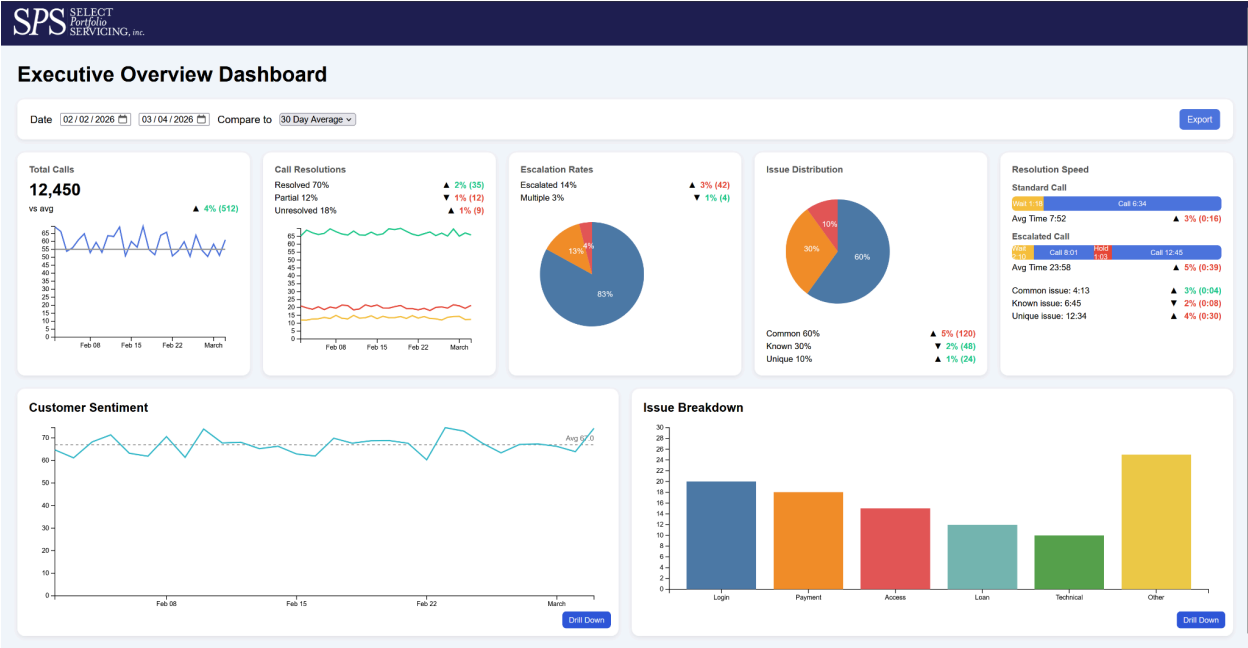
APPENDIX B: UI SKETCHES

Below is the structured list of views We have to design

Create mockups (Figma / PowerPoint / Adobe XD) matching these.

UI-1: Executive Overview Dashboard

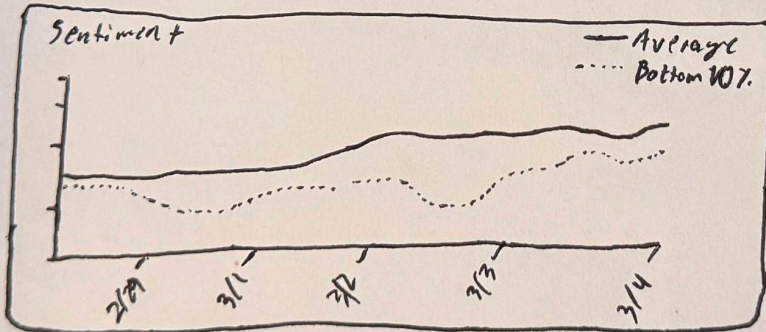
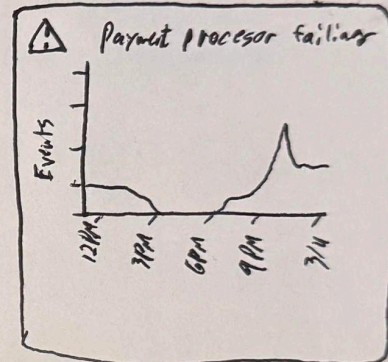
High-level performance snapshot.



UI-2: Daily Insights Pulse

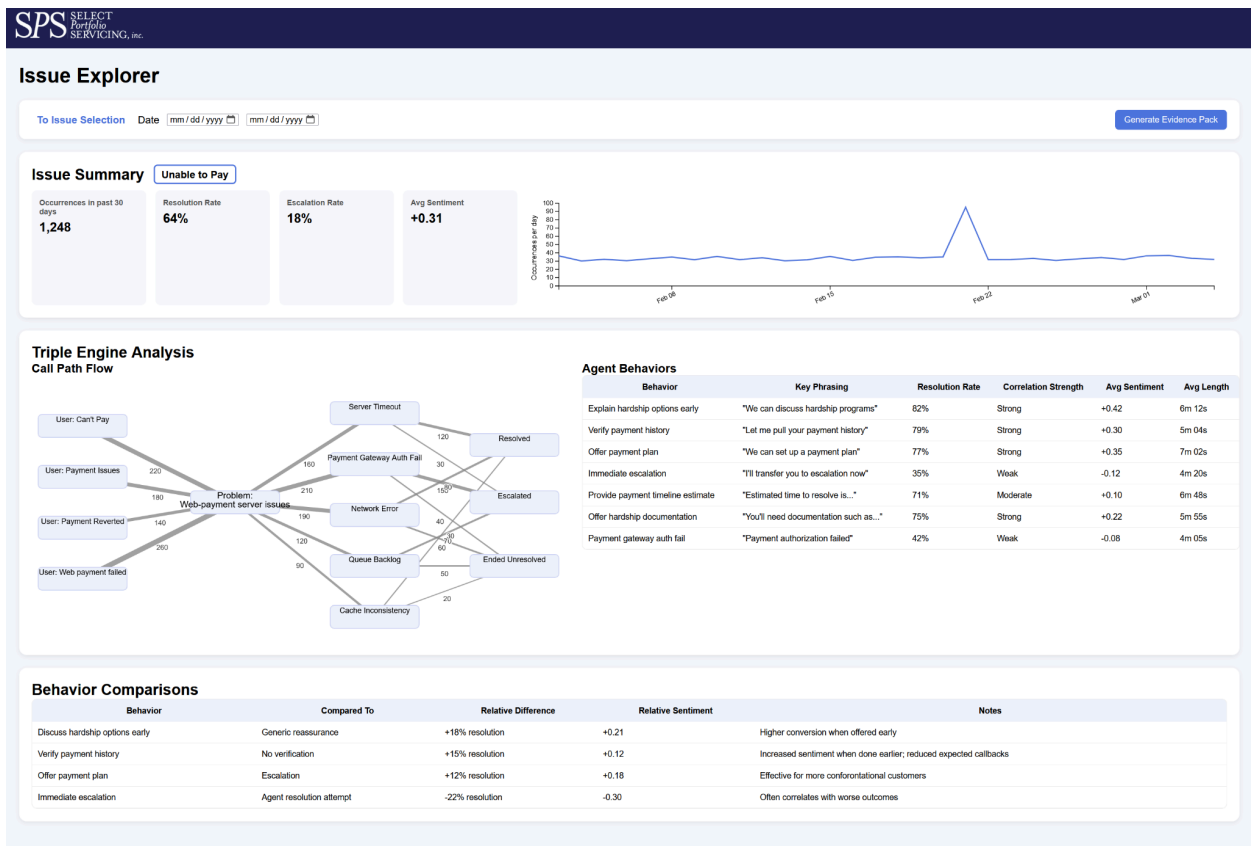
Daily Insight Pulse

Emerging Issues	#Occurrences	Severity	
Payment Processor failing	117	High	Evidence
Spanish translation incorrect	14	Moderate	Evidence
Website slow	9	Low	Evidence
...			

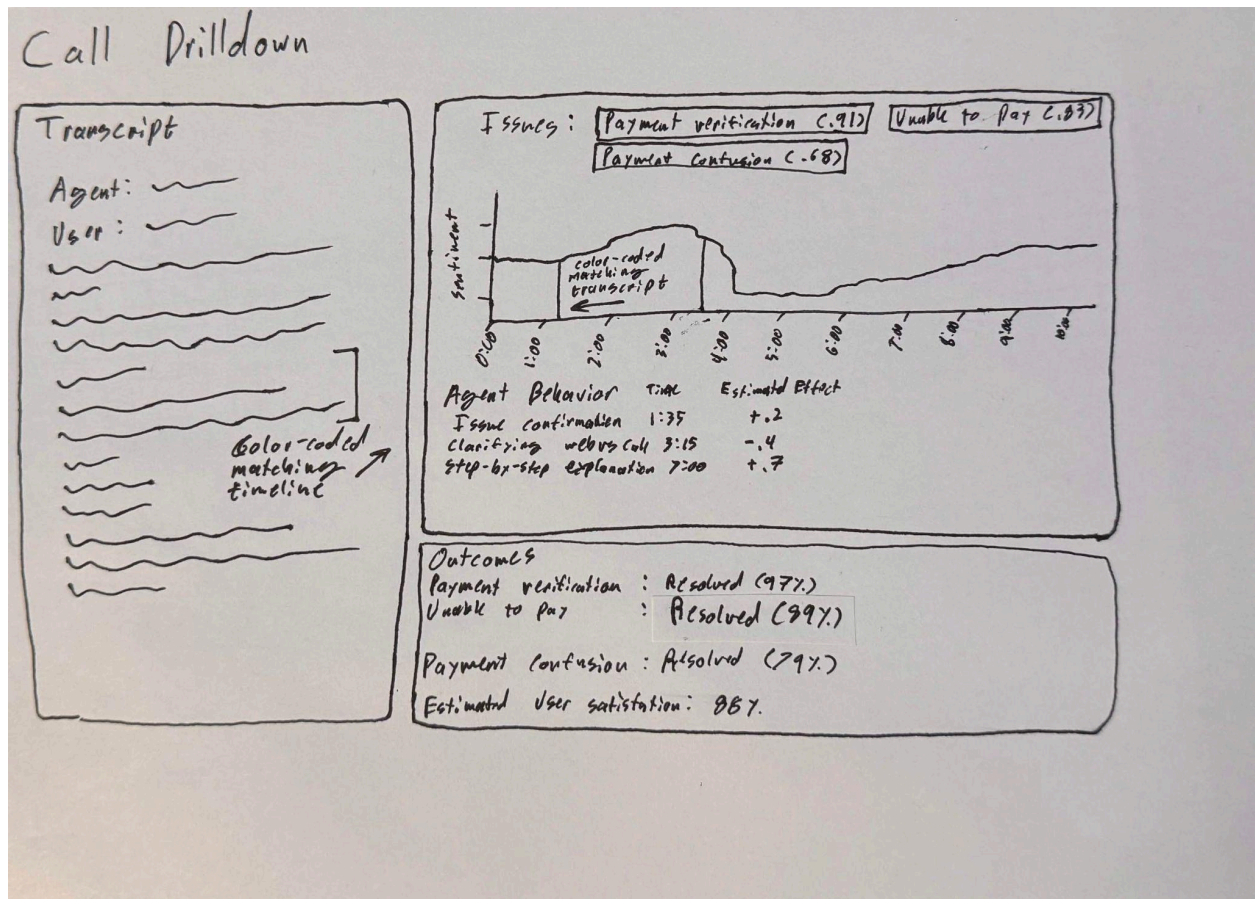


UI-3: Issue Explorer (Core Analytical Screen)

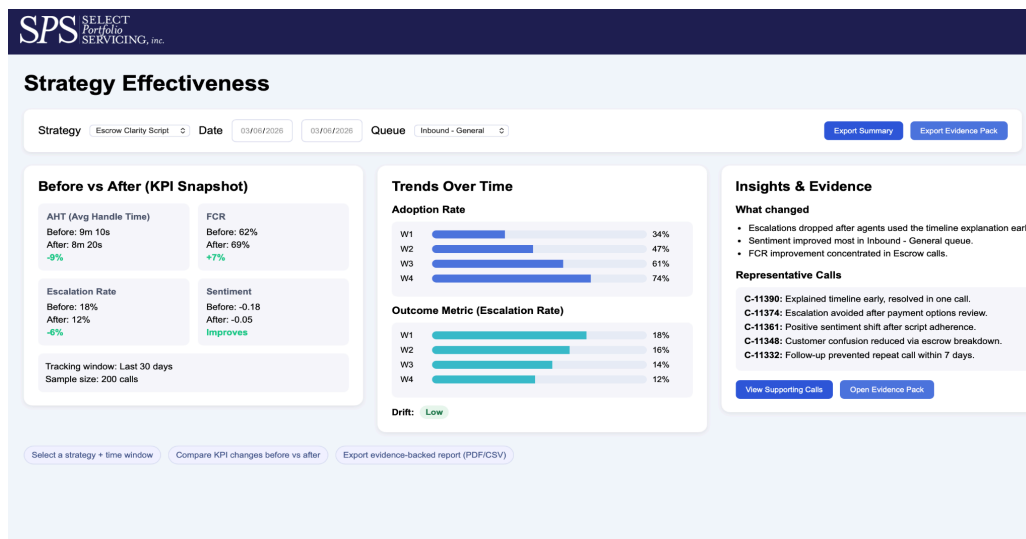
MOST IMPORTANT UI



UI-4: Call Drilldown View



UI-5: Strategy Effectiveness Dashboard



UI-6: Strategy Management

SPS

SELECT
Portfolio
SERVICING, inc.

Strategy Management

Date Range

03/27/2026

03/05/2026

Status

Active

Target Issue

Escrow

Owner/Team

Training

+ New Strategy

Export

Filters

Status

Active

Paused

Archived

Target Issue

Escrow

Owner/Team

Training

Date Window

Last 7 days

Last 30 days

Last 90 days

Reset Filters

Strategies

Strategy Name	Target Issue	Status	Adoption Rate	Outcome Change
Escrow Clarity Script	Escrow	Active	74%	+11%
Payment Plan Prompting	Payment	Active	68%	+8%
Complaint De-escalation Path	Complaint	Paused	41%	+2%
Hardship Options First	Payment	Active	82%	+13%
Escalation Deflection Pack	Complaint	Archived	57%	-1%
Escrow Timeline Coaching	Escrow	Active	63%	+6%
Structured Wrap-up Checklist	Complaint	Paused	49%	+3%

Filter strategies by status and issue

Track adoption and outcome changes

Detailed view of selected strategy

Selected Strategy Details

Description

Encourage agents to explain escrow changes in plain language, then confirm customer understanding before next-step recommendations.

Tracking Window

Last 30 days

Linked Evidence Pack

Evidence Pack 01

View Effectiveness

Edit Strategy

Strategies

Strategy Name	Target Issue	Status	Adoption Rate	Outcome Change
Escrow Clarity Script	Escrow	Active	74%	+11%
Payment Plan Prompting	Payment	Active	68%	+8%
Complaint De-escalation Path	Complaint	Paused	41%	+2%
Hardship Options First	Payment	Active	82%	+13%
Escalation Deflection Pack	Complaint	Archived	57%	-1%
Escrow Timeline Coaching	Escrow	Active	63%	+6%
Structured Wrap-up Checklist	Complaint	Paused	49%	+3%

Strategies

Strategy Name	Target Issue	Status	Adoption Rate	Outcome Change	Last Updated	Actions
Escrow Clarity Script	Escrow	Active	74%	+11%	2026-03-01	View Edit Archive
Payment Plan Prompting	Payment	Active	68%	+8%	2026-02-27	View Edit Archive
Complaint De-escalation Path	Complaint	Paused	41%	+2%	2026-02-24	View Edit Archive
Hardship Options First	Payment	Active	82%	+13%	2026-03-02	View Edit Archive
Escalation Deflection Pack	Complaint	Archived	57%	-1%	2026-02-11	View Edit Archive
Escrow Timeline Coaching	Escrow	Active	63%	+6%	2026-02-26	View Edit Archive
Structured Wrap-up Checklist	Complaint	Paused	49%	+3%	2026-02-20	View Edit Archive

CallInsights Design Document

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UI-7: Reports & Export

SPS SELECT
Portfolio
SERVICING, inc.

Reports & Export

Date

03/06/2026

03/06/2026

Issue Category

Escrow

Team

Training

Queue

Inbound - General

Sample Size (Top N)

20

Generate Evidence Pack

Download PDF

Export CSV

Evidence Pack Builder

Evidence Pack Contents

- ☒ Time window definition
- ☒ Filters applied (queue/issue/team)
- ☒ Counts & trends
- ☒ Sample size
- ☒ Representative call IDs
- ☒ Derived signals (sentiment shift)

UC-6 / CTO Export Notes

Builder includes export-ready sections for evidence pack generation and static PDF/CSV outputs.

Preview

Selected Filters

Date Range: 2026-02-01 to 2026-03-01
Filters applied: Queue = Inbound - General, Issue = Escrow, Team = QA
Sample Size: Top 20 calls

Key Metrics Snapshot

AHT 6m 42s	FCR 71%	Escalation Rate 17%	Avg Sentiment +0.26
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Representative Calls

Call ID	Issue	Team	Sentiment	Resolution	Notes
C-10981	Escrow	QA	+0.39	Resolved	Clear script adherence
C-10974	Payment	Ops	+0.18	Resolved	Plan accepted
C-10966	Complaint	Training	-0.05	Escalated	Supervisor callback
C-10952	Escrow	QA	+0.31	Resolved	Timeline clarified
C-10941	Payment	Ops	+0.24	Resolved	Late fee waived
C-10933	Escrow	QA	+0.22	Resolved	Transfer to specialist

UI-8: Compliance Dashboard

SPS SELECT
Portfolio
SERVICING, inc.

Compliance Monitoring Dashboard

From

02/05/2026

To

03/07/2026

Regulation

All

Search

Search transcripts...

Export Audit

Compliance Rate

60%

Percentage of reviewed calls that met required disclosures

Calls Reviewed

120

Total calls analyzed in selected range

Calls Missing Disclosures

48

Calls flagged for missing mandatory phrases

Top Missing Phrase

Disclose fee (17)

Most commonly omitted disclosure

Compliance Trend

● Compliant ● Non-compliant

Missing Phrase Distribution

Click a slice to filter calls

Non-compliant Calls

Showing calls missing required disclosures. Click "View" to inspect transcript.

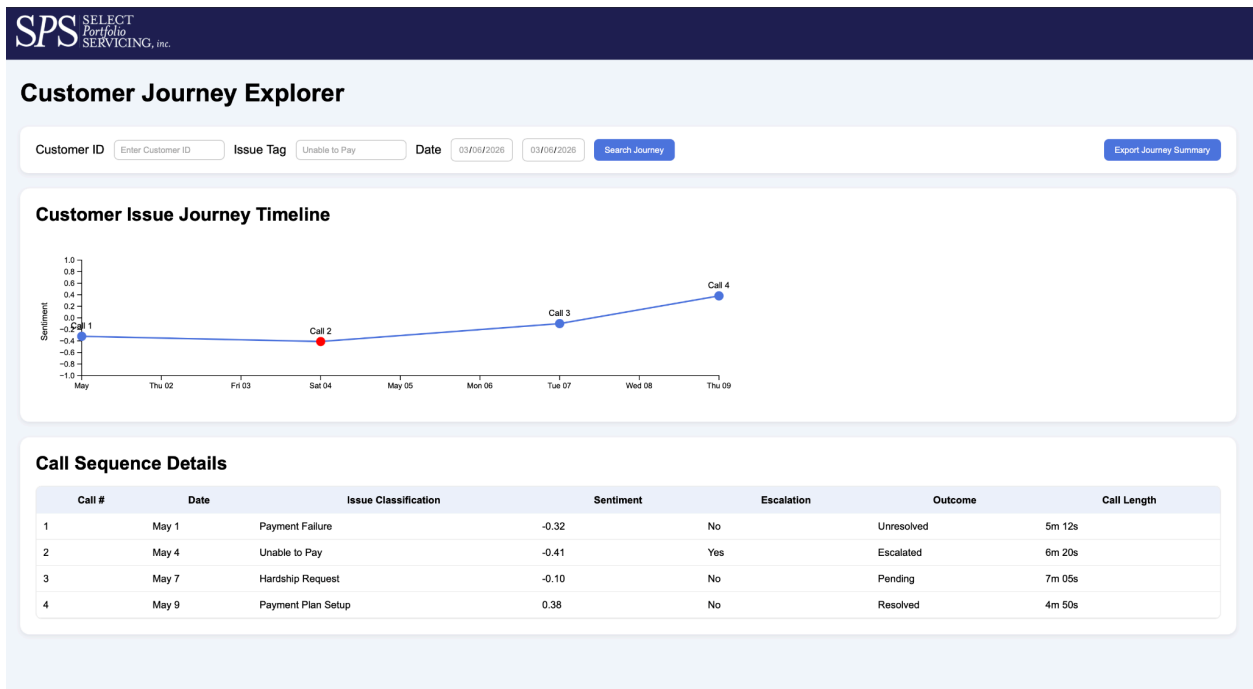
Call ID	Agent	Time	Missing Phrases	Actions
C1000	Bob	2026-02-13 00:49	Offer terms	View
C1001	Dana	2026-02-28 00:49	State rate	View
C1002	Bob	2026-03-01 00:49	State rate	View

Download CSV

CallInsights Design Document

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UI-9: Customer Journey Explorer

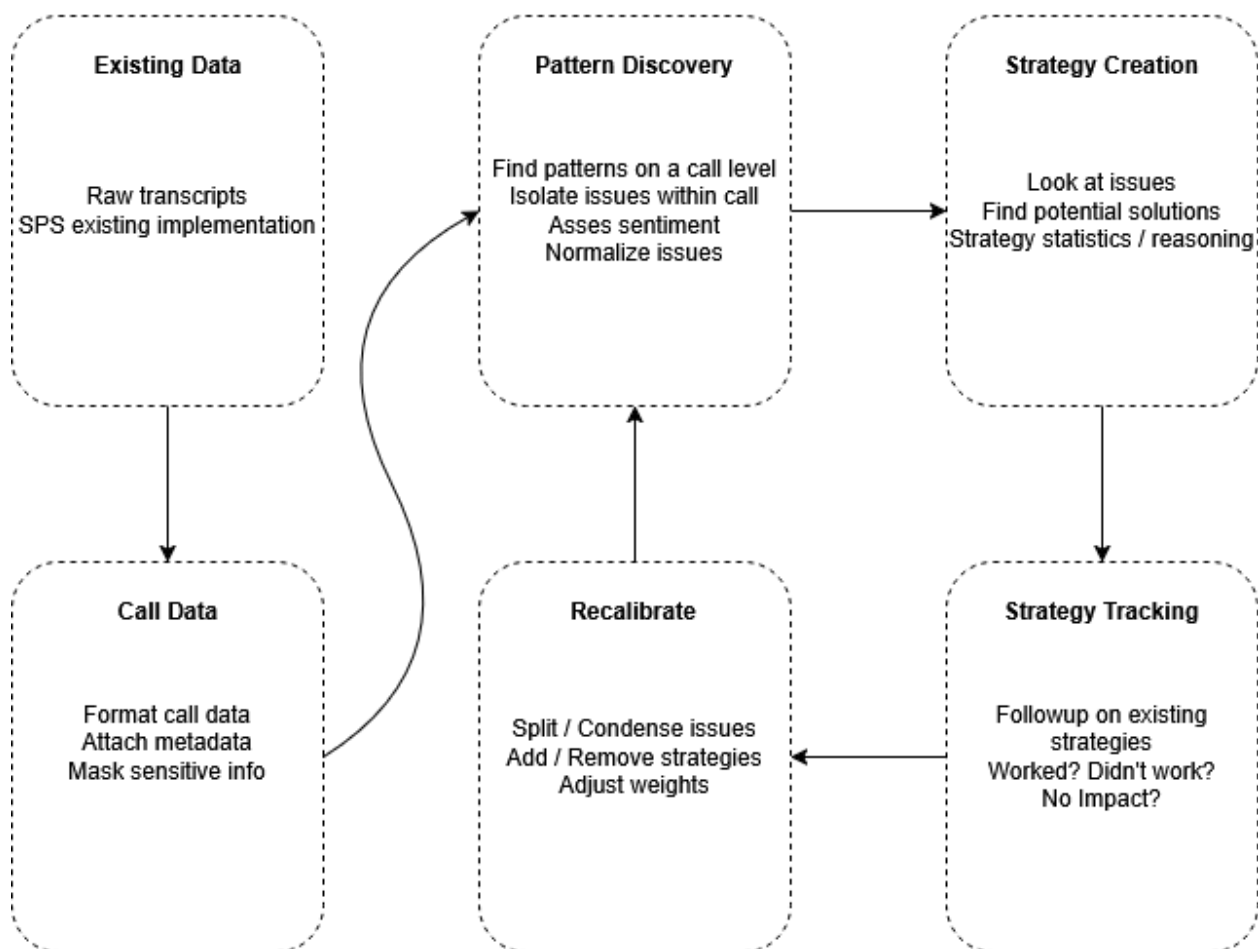


4. Requirements Analysis

4.1 System Architecture

The Call Insights (CI) platform is designed as a modular, web-based intelligence system that transforms raw call transcripts into actionable, evidence-backed operational strategies. The architecture follows a 7-layer intelligence framework that transforms raw call transcripts into evidence-backed operational strategies through AI analytics, behavioral modeling, and continuous learning. The system operates as a closed-loop intelligence cycle:

High Level Overview



Each layer is modular and communicates through structured APIs and a centralized data store, enabling scalability, extensibility, and clear separation of concerns.

Layer 1: Data Ingestion Layer

This layer is responsible for collecting and preparing raw call data.

Key responsibilities:

- Ingest call transcripts from internal sources
- Attach metadata (call ID, timestamp, agent ID)
- Mask sensitive customer information
- Store masked transcripts securely
- Store related information for traceability

This layer ensures that all incoming data is standardized and privacy-compliant before processing.

Layer 2: Transcript Processing Layer (Fact Gathering Engine)

This layer transforms raw transcripts into structured representations.

Key components include:

- **Call Segmentation Engine:** Splits calls into multiple issue-based segments
- **Transcript Structuring:** Identifies core questions:
 - Why did the customer call?
 - What did the agent do?
 - What did the agent ask the customer to do?
 - What was left unaddressed?
- **Sentiment Tracking:** Captures sentiment at opening, mid-call, and closing stages

Each segment is enriched with:

- Topic classification
- Sentiment trajectory
- Resolution status
- Agent behaviors
- Unaddressed issues

Layer 3: Behavioral & Outcome Extraction Layer

This layer extracts deeper semantic signals from each call segment.

Key functionalities:

- **Issue Classification:** Using topic modeling, clustering, and supervised learning
- **Agent Behavior Detection:** Identifies actions such as empathy, escalation, or interruptions
- **Customer Action Extraction:** Determines instructions given to customers

- **Unaddressed Issue Detection:** Identifies gaps leading to repeat calls
- **Outcome Classification:** Labels segments as resolved/unresolved, escalated, and sentiment-based

All outputs are structured and stored for analysis.

Layer 4: Analytical Intelligence Layer

This is the core analytics engine that identifies patterns across large datasets.

Key components:

- **Triple Engine Analysis:**
Correlates:
 - Customer Problem
 - Agent Behavior
 - Outcome
- **Statistical Analysis:**
Computes correlation strength, lift, and significance
- **Trend Detection:**
Identifies:
 - Emerging issues
 - Volume spikes
 - Sentiment changes
 - Strategy drift
- **Knowledge Graph:**
Visualizes relationships between issues, behaviors, and outcomes
- **Alerting System:**
Detects anomalies and generates alerts

Layer 5: Evidence & Reporting Layer

This layer ensures transparency and explainability of insights.

Evidence Pack includes:

- Time window and filters
- Sample size
- Metrics and trends
- Representative call IDs
- Derived analytical signals

This enables users to validate insights with real data.

Layer 6: Strategy & Continuous Learning Layer

This layer converts insights into actionable improvements.

Key components:

- **Strategy Creation:** Define operational strategies
- **Lifecycle Management:** Track strategies through stages:
 - Proposed -> Accepted -> Implemented -> Evaluating -> Closed
- **Effectiveness Tracking:** Measure impact using before/after comparisons

This layer enables continuous improvement through data-driven decisions.

Layer 7: Presentation & Governance Layer (Web Application)

CI is delivered as a unified web application, not a collection of dashboards.

Key user-facing modules:

- Executive Overview
- Daily Insights Pulse
- Monthly Recalibration
- Issue Explorer
- Call Drilldown
- Strategy Management
- Strategy Effectiveness
- Reports & Export
- Sentiment Journey Map
- Voice-to-Insight
- Natural Language Query (Ask CI)
- Admin & AI Governance

This layer provides an interactive workspace where users explore insights, validate findings, and take action.

Layer 8: Pre-Call Pattern Layer (Dynamic IVR Intelligence)

This layer is implemented after sufficient learning has been achieved.

Inputs:

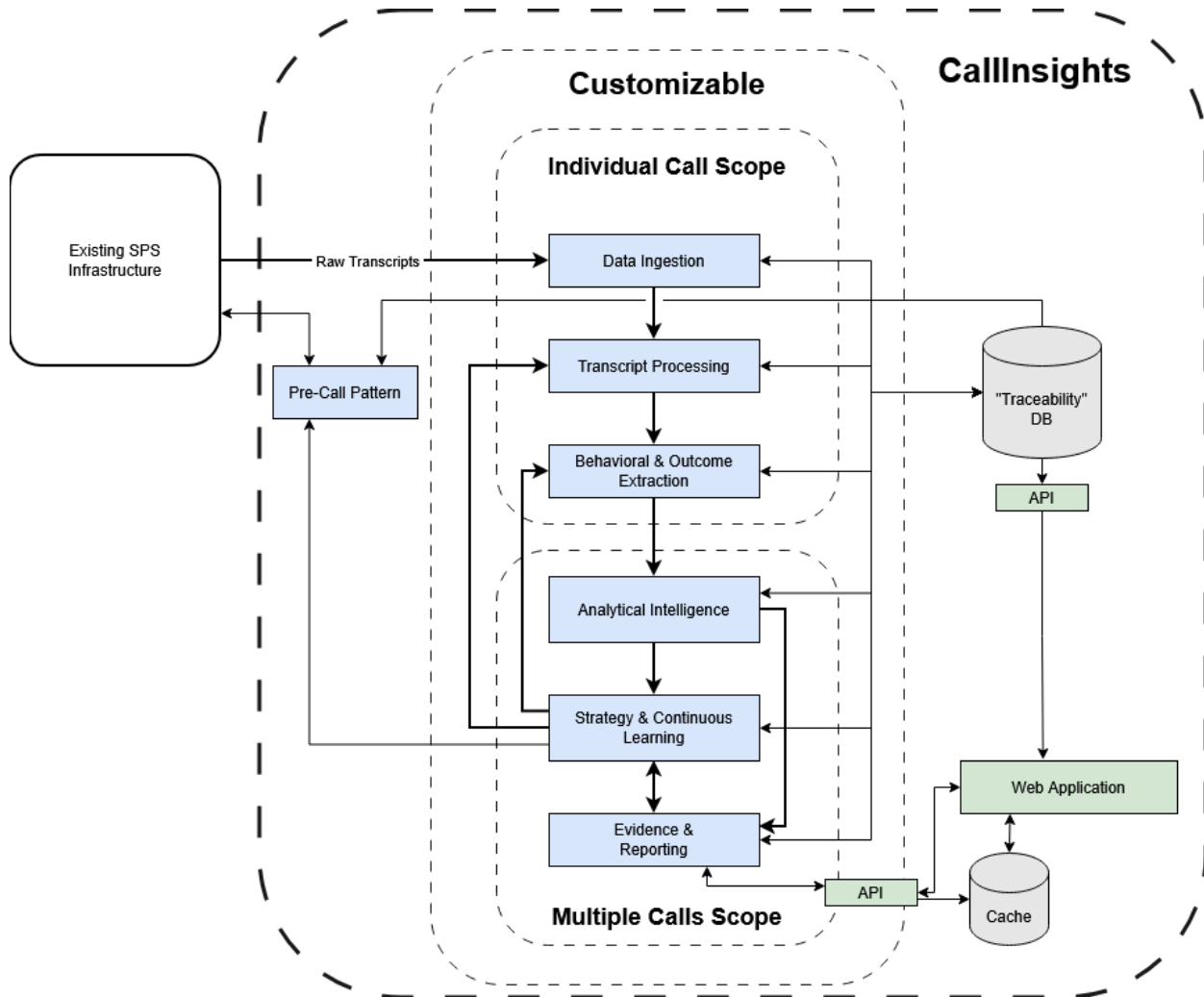
- Recurring issues
- Call intent distributions
- Successful resolution patterns

Outputs:

- Smarter IVR routing

- Issue-aware call tagging
- Early metadata enrichment

This layer would extend CI to be capable of upstream call handling.



Cache will primarily handle automated reports

4.2 Personnel

The development of the Call Insights (CI) platform is distributed across all team members to ensure balanced contribution across the full 7-layer intelligence framework. Rather than assigning ownership of entire layers, responsibilities are divided by subsystem and technical specialization, with each member contributing to multiple layers of the architecture.

Abdulrahman Al Hattali: Backend Infrastructure & Data Pipeline Lead

Abdulrahman is responsible for designing and implementing the core backend infrastructure that supports data flow across the CI platform.

Primary Responsibilities:

- Data Ingestion Layer (Layer 1)
 - Transcript ingestion pipeline
 - Metadata attachment and storage
 - Data masking and privacy compliance
- Backend API development for inter-layer communication
- Database schema design for transcripts, segments, and derived signals
- Integration between processing layers and frontend services

Cross-Layer Contributions:

- Supports Transcript Processing Layer (Layer 2) by optimizing data pipelines
- Enables Evidence & Reporting Layer (Layer 5) through efficient data retrieval

Justification:

Abdulrahman has strong experience in backend development and data systems, making him well-suited for handling data pipelines, APIs, and scalable infrastructure.

Alex Held - Frontend & Web Application Lead

Alex is responsible for building the unified CI web application and ensuring a seamless, interactive user experience across all presentation modules.

Primary Responsibilities:

- Presentation & Governance Layer (Layer 7)
 - Core web application architecture
 - Navigation and interaction design
 - Implementation of all UI modules:
 - Executive Overview
 - Issue Explorer
 - Call Drilldown
 - Strategy Management
- Integration of frontend with backend APIs

Cross-Layer Contributions:

- Works with Analytical Layer (Layer 4) to visualize insights
- Supports Strategy Layer (Layer 6) through interactive UI tools
- Contributes to Voice-to-Insight and Ask CI interfaces

Justification:

Alex has experience in frontend development and UI/UX design, making him well-equipped to create an intuitive and visually engaging web application.

Anshul Kasera - Analytics & Intelligence Engine Lead

Anshul is responsible for implementing the core analytical, statistical, and AI-driven components that power CI's intelligence capabilities, including contributions to NLP system development and integration.

Primary Responsibilities:**Analytical Intelligence Layer (Layer 4)**

- Triple Engine (Problem, Behavior, Outcome correlation)
- Statistical computations (correlation, lift, significance)
- Trend detection (spikes, anomalies, drift)
- Knowledge Graph construction and analysis
- Alerting system for anomaly detection

AI & NLP Implementation Contributions (Layer 2 and Layer 3)

- Co-implementation of NLP pipelines for transcript processing and feature extraction
- Integration of sentiment analysis, issue classification, and outcome detection outputs into the analytics engine
- Development of models that leverage both structured data and NLP-derived features
- Feature engineering from text-based and structured data sources
- Model evaluation, performance tuning, and validation across AI components

Cross-Layer Contributions:

- Supports Behavioral & Outcome Extraction Layer (Layer 3) through both model development and NLP integration
- Enables Strategy Layer (Layer 6) through insight generation
- Provides data for Evidence Packs (Layer 5)
- Ensures seamless flow of NLP-derived signals into downstream intelligence systems

Justification:

Anshul's background in machine learning, statistical analysis, NLP and data modeling enables him to contribute not only to CI's analytical backbone but also to the implementation and integration of NLP-driven intelligence within the platform.

Prachi Aswani - NLP, AI Systems & Product Integration Lead

Prachi is responsible for the natural language processing components and advanced AI-driven features that enable CI's deep understanding of call transcripts and user interaction.

Primary Responsibilities:

- Transcript Processing Layer (Layer 2)
 - Call segmentation
 - Sentiment analysis over time
- Behavioral & Outcome Extraction Layer (Layer 3)
 - Issue classification
 - Agent behavior detection
 - Outcome classification
- AI-driven features:
 - Natural Language Query (Ask CI)
 - Voice-to-Insight pipeline

Cross-Layer Contributions:

- Supports Analytical Layer (Layer 4) by providing structured NLP outputs
- Enhances Presentation Layer (Layer 7) with intelligent interaction features
- Contributes to Pre-Call Pattern Layer (Layer 8) through learned patterns

Justification:

Prachi has strong experience in machine learning, NLP, and data analysis, along with prior work on large-scale text classification systems, making her highly qualified to lead CI's AI and language processing components.

Team Collaboration Model

All team members collaborate across layers to ensure integration and system coherence. Key collaborative areas include:

- End-to-end pipeline integration (Layer 1 to Layer 7)
- Evidence Pack generation and validation
- Strategy tracking and evaluation workflows
- Advanced features (Rank 3), including AR/VR visualization and simulation tools

This distributed ownership model ensures that:

- Each team member contributes to multiple parts of the system
- No single point of failure exists
- The system remains modular, scalable, and well-integrated

4.3 System Features

The CI platform features are categorized into three priority levels.

High-Level Feature Grid

Feature Area	Priority	Feature	Description
Data	1	Transcript Ingestion	Upload and store transcripts with metadata and masking
Processing	1	Call Segmentation	Split calls into multi-issue segments
NLP	1	Issue Classification	Identify why customers called
NLP	1	Behavior Extraction	Detecting agent actions (empathy, escalation, etc.)
NLP	1	Sentiment Analysis	Track sentiment over time
NLP	1	Outcome Classification	Label resolution and escalation status
Analytics	1	Triple Engine	Correlate problem, behavior, and outcome
UI	1	Core Web Interface	Navigation and basic data viewing

Feature Area	Priority	Feature	Description
Insights	2	Issue Explorer	Interactive analysis of issue patterns
Insights	2	Call Drilldown	View transcripts with extracted signals
Insights	2	Daily Insights Feed	Show emerging issues and anomalies
Strategy	2	Strategy Management	Create and track strategies
Strategy	2	Effectiveness Tracking	Measure impact of strategies
Reporting	2	Evidence Pack Export	Generate structured reports
AI	2	Natural Language Query (Ask CI)	Query system using plain English
Voice	2	Voice-to-Insight	Convert speech to analyzed transcript
Alerts	2	Smart Alerts	Notify users of anomalies

Feature Area	Priority	Feature	Description
Visualization	3	Knowledge Graph	Visualize relationships between variables
AI	3	Predictive Analytics	Predict escalation and repeat calls
AI	3	Strategy Recommendations	Suggest optimal strategies
Simulation	3	Call Simulation Tool	Simulate real-time call analysis
XR	3	AR/VR Insight Visualization	Explore insights in immersive environments
Collaboration	3	Insight Annotation	Team-based commenting and tagging
Explainability	3	AI Transparency Panel	Show reasoning behind insights